ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 8 strands: Life and Living (photosynthesis and respiration compared, biotic and abiotic factors, a predator–prey graph, decomposers, food spoilage and preservation), Matter and Materials (diffusion, density and floating, atoms and the periodic table, metals reacting with acid), Energy and Change (resistance, the heating and magnetic effects of current, fuses, shadows and coloured light) and Planet Earth and Beyond (planet temperatures, telescopes in South Africa, light-years and the Southern Cross).

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL calculations with formulae and units. A non-programmable calculator may be used.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which gas do plants give off during photosynthesis?

A Carbon dioxide B Nitrogen C Oxygen D Hydrogen

1.2 Which of the following is an ABIOTIC factor in an ecosystem?

A Grass B Sunlight C Bacteria D Insects

1.3 Organisms that break down dead plants and animals are called …

A decomposers. B producers. C herbivores. D carnivores.

1.4 Keeping food in a fridge slows down spoilage because micro-organisms …

A are killed by the cold. B need light to grow.
C grow very slowly when it is cold. D cannot live in closed containers.

1.5 The spreading of particles from where there are many to where there are few is called …

A diffusion. B condensation. C expansion. D melting.

1.6 The positively charged particles in the nucleus of an atom are …

A electrons. B neutrons. C molecules. D protons.

1.7 When magnesium reacts with dilute hydrochloric acid, the gas given off is …

A oxygen. B hydrogen. C carbon dioxide. D chlorine.

1.8 Which device melts and breaks the circuit when too much current flows?

A Switch B Resistor C Fuse D Ammeter

1.9 A red T-shirt is seen in a room lit only by green light. The T-shirt will look …

A red. B green. C white. D black.

1.10 The distance that light travels in one year is called a …

A light-year. B galaxy. C solar year. D orbit.

(10)

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number. If the statement is FALSE, write down the correct word(s) to replace the underlined word(s).

2.1 Respiration takes place in <u>all living cells</u>, day and night. (2)

2.2 An object with a density of 1,5 g/cm³ will <u>float</u> in water. (2)

2.3 A <u>thicker</u> wire has a lower resistance than a thinner wire of the same length and material. (2)

2.4 Shadows form because light travels in <u>curved</u> lines. (2)

2.5 MeerKAT in the Northern Cape is a <u>radio</u> telescope. (2)

(10)

TOTAL SECTION A: [20]

SECTION B

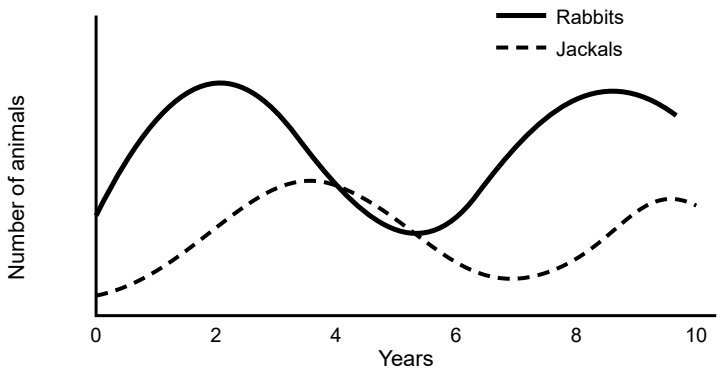
QUESTION 3: LIFE AND LIVING

3.1 Copy and complete the table to compare photosynthesis and respiration. (4)

	Photosynthesis	**Respiration**
Where it takes place		
Gas taken in		

3.2 Give TWO uses of the glucose that a plant makes. (2)

3.3 The graph below shows the numbers of rabbits and jackals in a nature reserve over 10 years.



Numbers of rabbits and jackals

3.3.1 Which animal is the predator? (1)

3.3.2 Explain why the number of jackals rises soon after the number of rabbits rises. (2)

3.3.3 Explain why the number of rabbits then falls. (2)

3.4 Explain why decomposers are important in an ecosystem. (2)

3.5 Name THREE conditions that micro-organisms need to grow on food. (3)

3.6 Name TWO methods of preserving food and explain how ONE of them stops micro-organisms from spoiling the food. (4)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 A purple crystal of potassium permanganate is placed carefully at the bottom of a beaker of still water. After an hour, the whole beaker of water is pale purple.

4.1.1 Name the process that takes place. (1)

4.1.2 Use the particle model to explain what happens. (2)

4.1.3 Would this happen faster or slower in hot water? Give a reason. (2)

4.2 A metal block has a mass of 54 g and a volume of 20 cm³.

4.2.1 Calculate the density of the block. (3)

4.2.2 The density of water is 1 g/cm³. Will the block float or sink in water? Give a reason. (2)

4.3 An atom of carbon has 6 protons, 6 neutrons and 6 electrons.

4.3.1 Where in the atom are the protons and neutrons found? (1)

4.3.2 What is the overall charge of the atom? Explain. (2)

4.4 Give TWO properties of metals and TWO properties of non-metals. (4)

4.5 A learner adds a piece of zinc to dilute sulfuric acid in a test tube.

4.5.1 Describe what the learner will observe. (1)

4.5.2 Describe a test for the gas that is formed, and give the result. (2)

[20]

QUESTION 5: ENERGY AND CHANGE

5.1 Give TWO factors that affect the resistance of a wire. (2)

5.2 Name the effect of electric current that is used in EACH of the following: (a) a kettle; (b) an electric bell; (c) a light bulb. (3)

5.3 Explain how to make a simple electromagnet, and give ONE way to make it stronger. (3)

5.4 Explain why a fuse must be connected in series with an appliance. (2)

5.5 A torch shines on a ball in front of a wall. Draw a labelled ray diagram to show how the shadow of the ball forms on the wall. (3)

5.6 Explain why a white shirt looks white in sunlight, but a black shirt looks black. (4)

5.7 Light travels at 300 000 km/s. Explain why we see lightning before we hear the thunder. (3)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1 Study the table of planet data.

Planet	Mercury	Venus	Earth	Mars

Average distance from the Sun (million km)	58	108	150	228
Average surface temperature (°C)	167	464	15	−65

6.1.1 Which planet is the hottest? (1)

6.1.2 Mercury is closer to the Sun than Venus, but Venus is hotter. Suggest a reason. (2)

6.1.3 Describe the general pattern between distance from the Sun and temperature. (2)

6.2 Name the large optical telescope near Sutherland in the Northern Cape. (1)

6.3 Explain the difference between an optical telescope and a radio telescope. (2)

6.4 A star is 4 light-years away. Explain what this means. (2)

6.5 Describe the shape of the Milky Way galaxy, and state where our solar system is in it. (3)

6.6 Explain how the Southern Cross can be used to find south at night. (3)

6.7 Why do we see the Moon change shape (phases) during a month? (2)

6.8 Is the Moon a planet, a star or a natural satellite? (1)

6.9 Give ONE reason why the Moon has many craters. (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER